US008996698B1

(12) **United States Patent**
Tagg

(10) **Patent No.: US 8,996,698 B1**
(45) **Date of Patent: *Mar. 31, 2015**

(54) **COOPERATIVE NETWORK FOR MOBILE INTERNET ACCESS**

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(73) Assignee: **Truphone Limited** (GB)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 1268 days.

This patent is subject to a terminal disclaimer.

(21) Appl. No.: **09/706,406**

(22) Filed: **Nov. 3, 2000**

(51) **Int. Cl.**
G06F 15/16 (2006.01)

(52) **U.S. Cl.**
USPC **709/226**; 709/204; 709/229

(58) **Field of Classification Search**
USPC 709/227, 229, 225; 455/428, 430, 433, 455/436, 414.1; 379/114.2; 370/401, 353, 370/235; 726/21, 25; 717/118; 706/45
See application file for complete search history.

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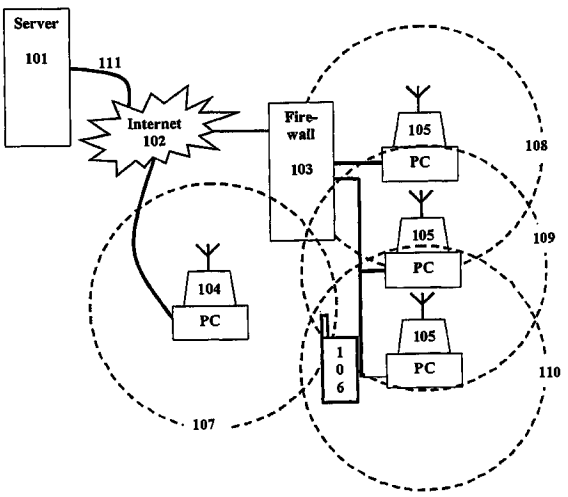
Primary Examiner — Douglas Blair

(57) **ABSTRACT**

A method and system for connecting mobile users to the Internet using cooperative interconnection. A node consists of a network element that has two connection methodologies—typically a node will have a radio link and a wired link to the Internet. To obtain a connection a member of the service locates themselves in proximity to a network node which is commonly provided by another member and the cooperative tunneling agent requests a secure link. All members of the service agree to give connection to the Internet upon request from another member.

13 Claims, 9 Drawing Sheets



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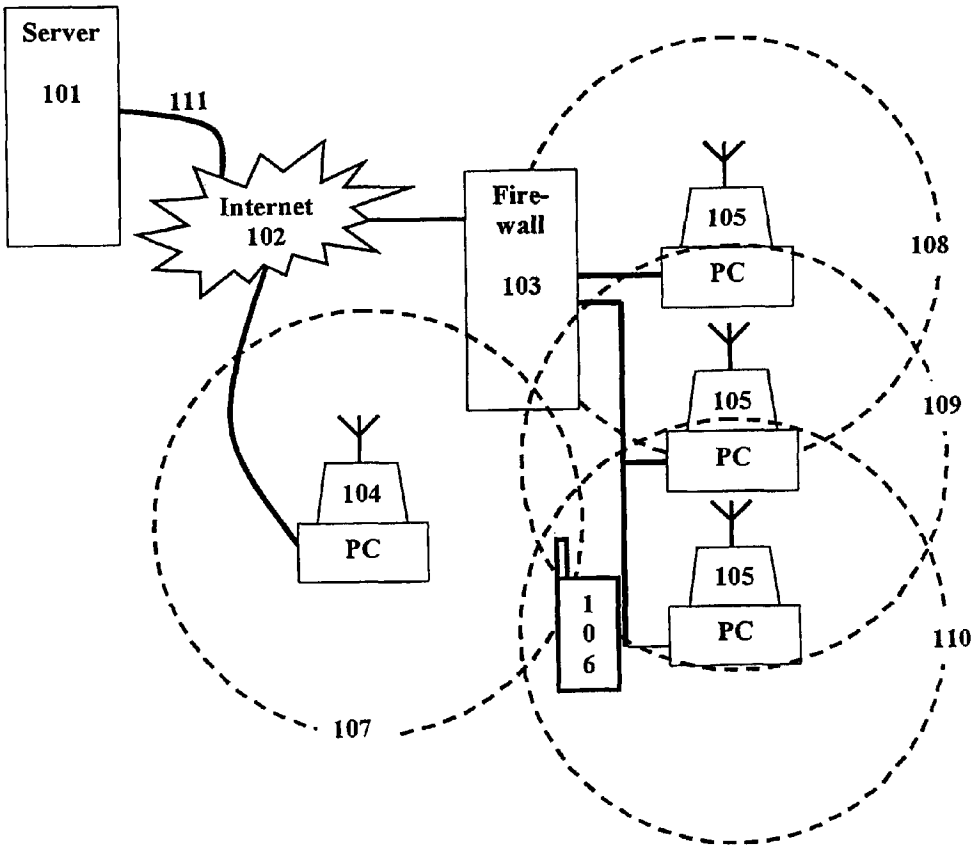
Server
101
111
Internet
102
Fire-
wall
103
105
PC
108
105
PC
109
104
PC
105
PC
1
0
6
110
107

Fig. 1

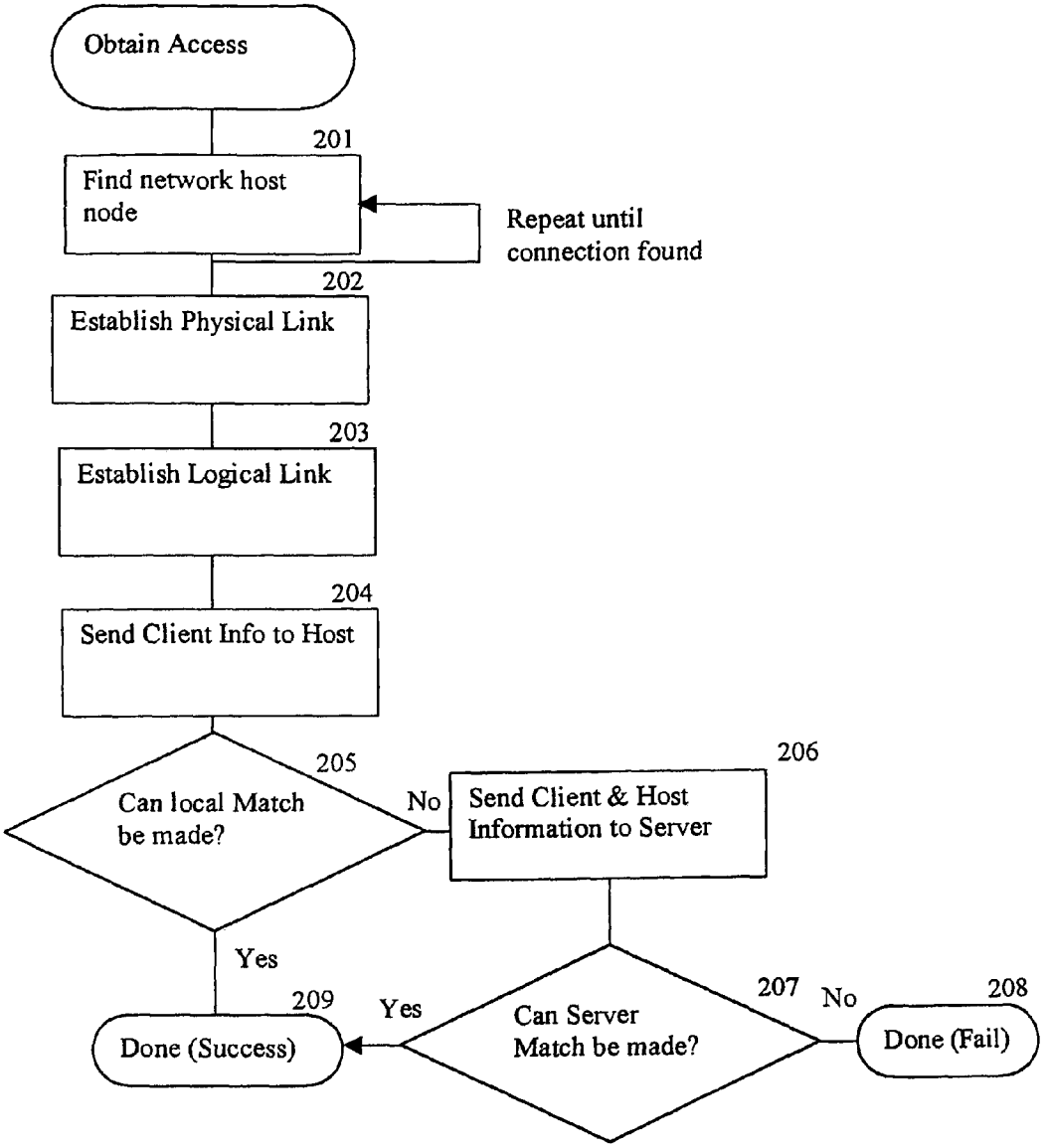
Obtain Access
201
Find network host node
Repeat until connection found
202
Establish Physical Link
203
Establish Logical Link
204
Send Client Info to Host
205
Can local Match be made?
No
206
Send Client & Host Information to Server
Yes
209
Done (Success)
Yes
207
Can Server Match be made?
No
208
Done (Fail)

Fig. 2

General Member Information 301

Member ID	#
Customer Record	Name, Address etc..

Tunnel Provider	**302**
Allow data link	YES/NO
Allow voice link	YES/NO
Bandwidth share	5%-ALL
Number of simultaneous users	1-16
Number of provided minutes	n
Number of discrete connection events	n
Total bits provided	n
Charge for uplink	$/hour
Cost to establish connection	$per connect

Tunnel user	**303**
Encryption level (data)	low-high
Encryption level (voice)	low-high
Pay for up-link	max $/hour
Number of used minutes (budget)	n
Link budget	max $/day
Polling Frequency	per hour
Cell phone record	Carrier, number etc...
Synchronize or buy	Table of desired content

Fig. 3

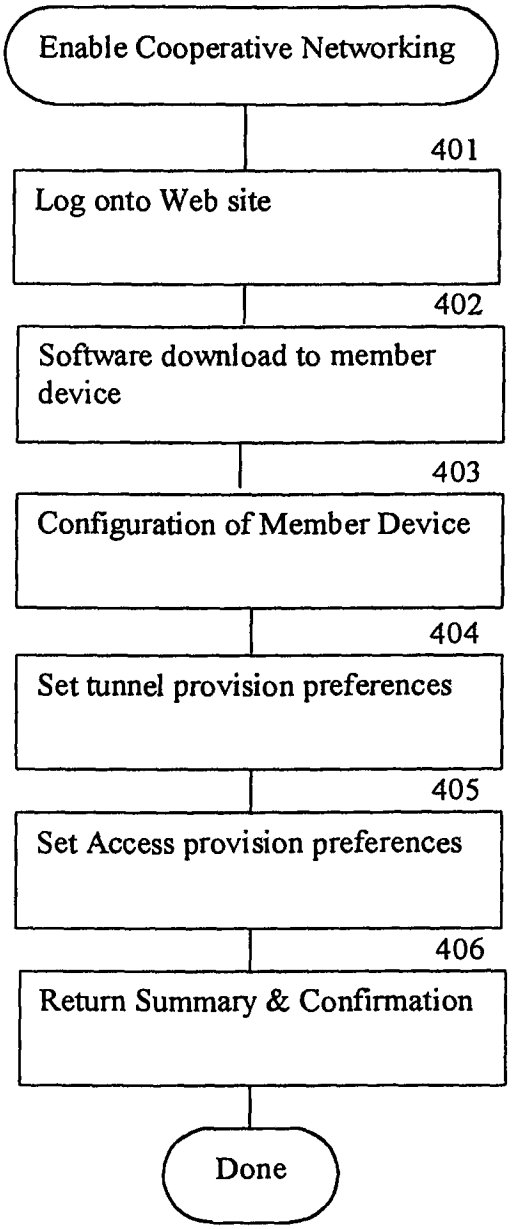
Enable Cooperative Networking
401
Log onto Web site
402
Software download to member device
403
Configuration of Member Device
404
Set tunnel provision preferences
405
Set Access provision preferences
406
Return Summary & Confirmation
Done

Fig. 4

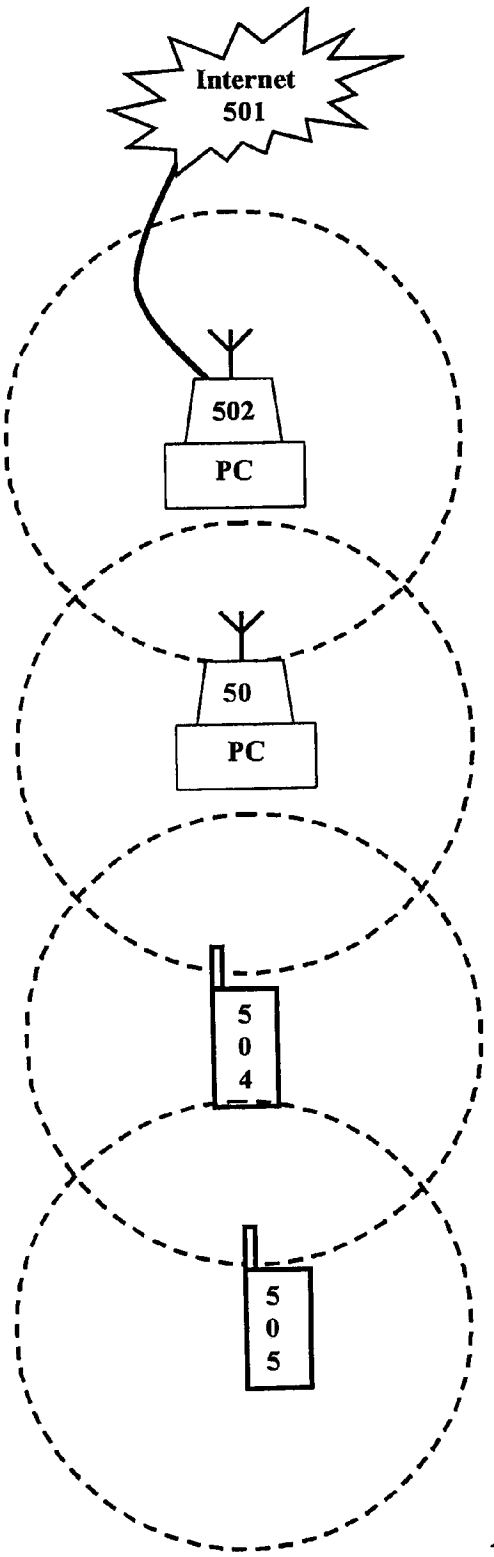

Fig. 5

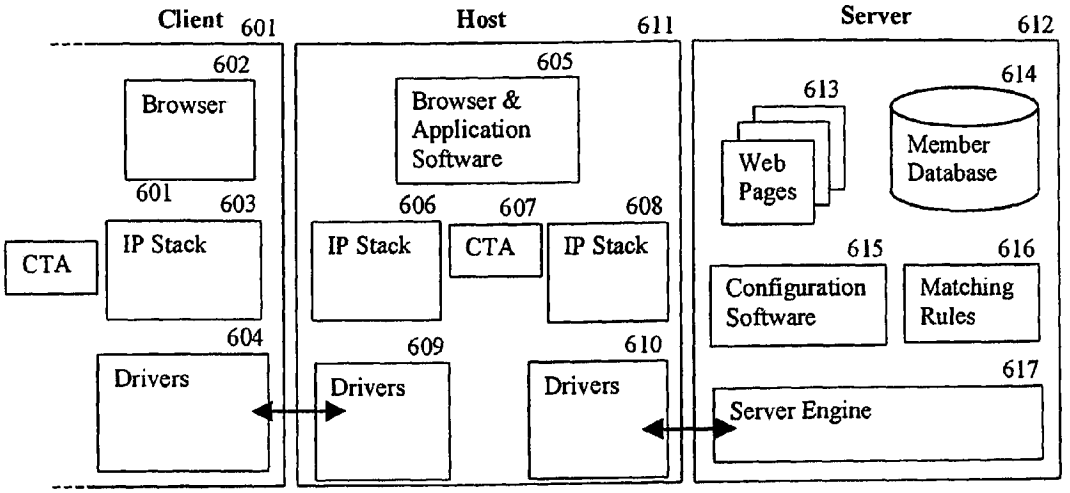
Client 601
Host 611
Server 612
602
Browser
605
Browser & Application Software
613
614
Web Pages
Member Database
601
603
606
607
608
CTA
IP Stack
IP Stack
CTA
IP Stack
615
616
Configuration Software
Matching Rules
604
609
610
617
Drivers
Drivers
Drivers
Server Engine

Fig. 6

Facebook's Exhibit No. 1058
Page 9

	Cable	Cell phone Slow Move	Cellphone Mobile pos	LAN (T1)	802.11	Bluetooth	LAN (T1)	Cell Phone Slow Move	Cell phone Fast Move	Bluetooth	ADSL + Their LAN	Cell Phone	Cable
Today	500	9.6	9.6	10M	11M	1M	10M	9.6	9.6	1M	256	9.6	500
Min	100	28	28	10M	1M	1M	10M	28	28	1M	128	28	100
Max	2000	128	56	100M	11M	10M	100M	128	56	10M	2000	128	2000
	Home	Walking to train	In Train	In Office	In Meeting	Lunch Starbucks	Office Desk	Walk to Car	Car	Gas Station	Friend's Home	Taxi	Hotel
	701	702	703	704	705	706	707	708	709	710	711	712	713

Fig. 7

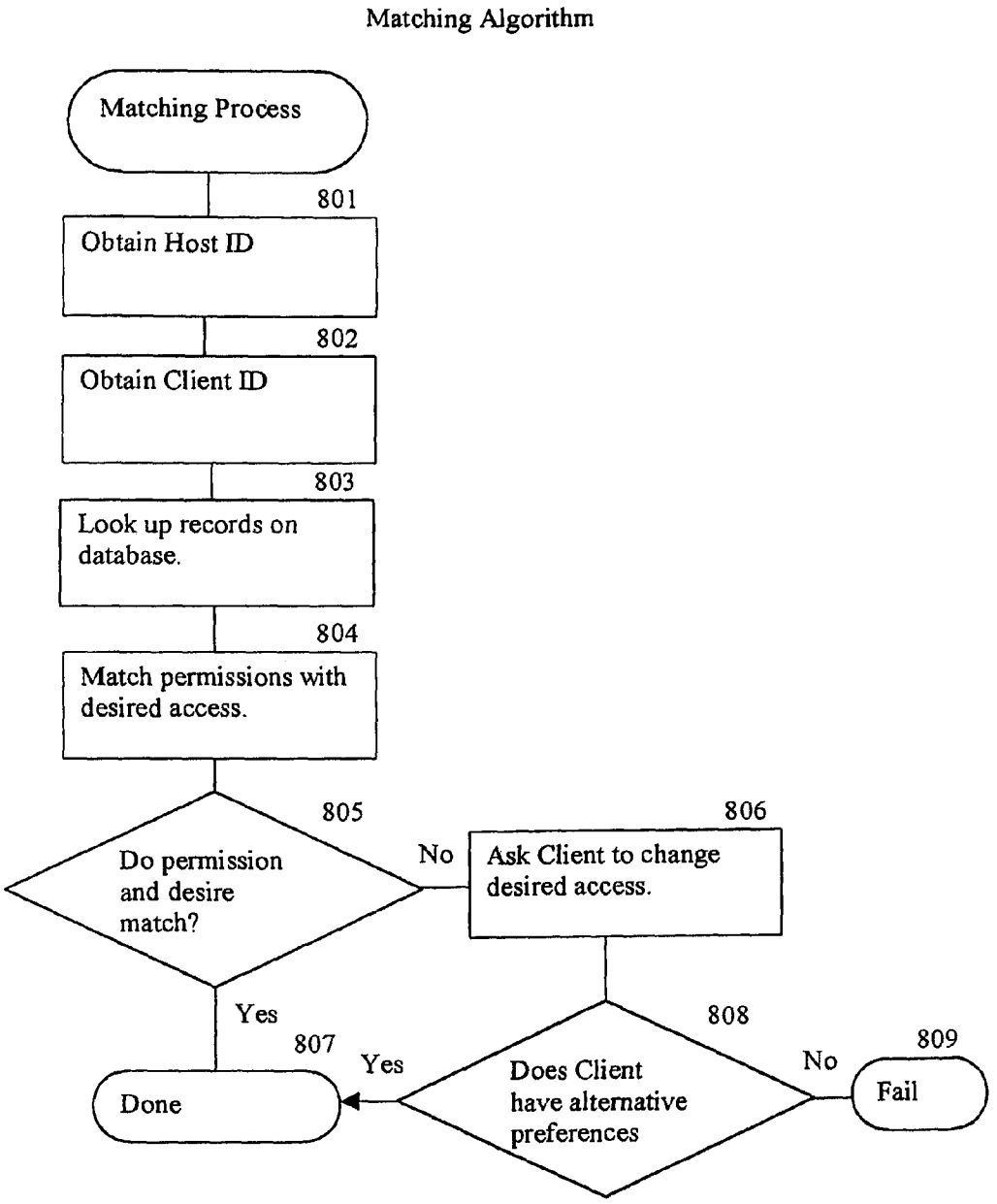
Matching Algorithm
Matching Process
801
Obtain Host ID
802
Obtain Client ID
803
Look up records on database.
804
Match permissions with desired access.
805
Do permission and desire match?
No
806
Ask Client to change desired access.
Yes
807
Done
Yes
808
Does Client have alternative preferences
No
809
Fail

Fig. 8

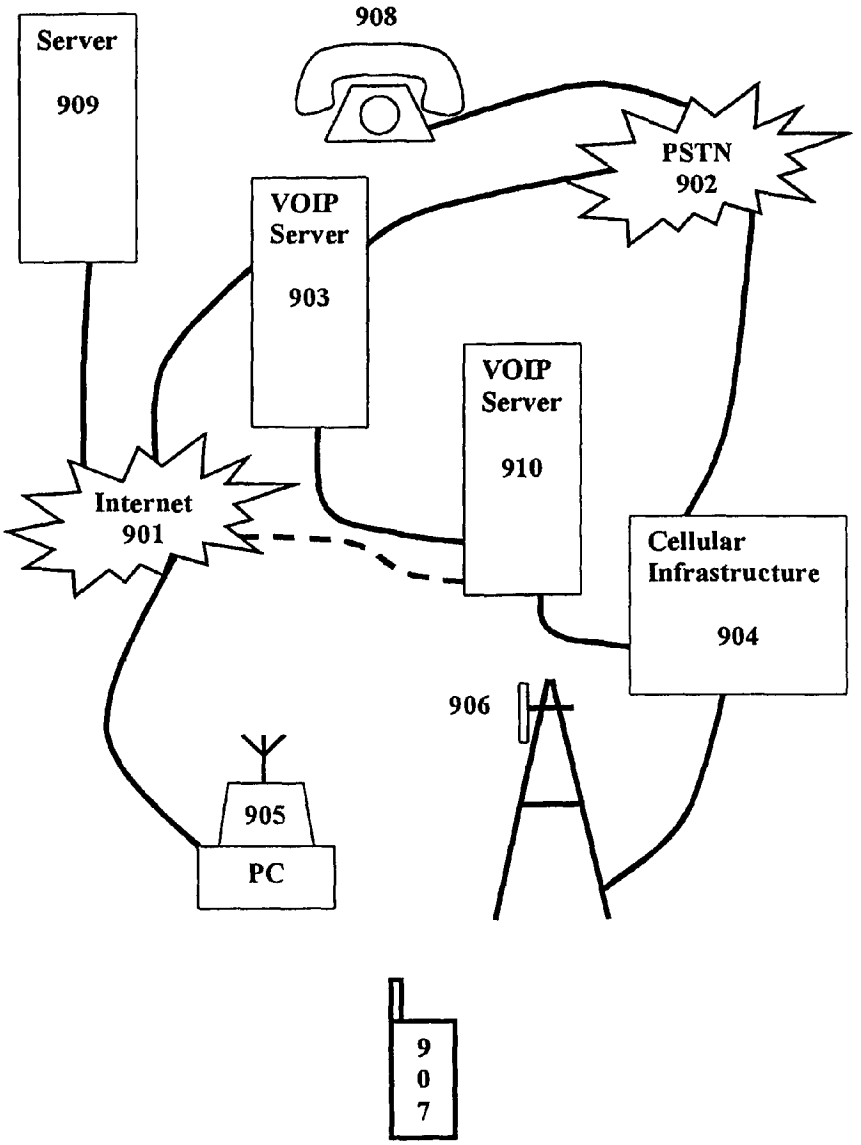
Server
909
908
PSTN
902
VOIP
Server
903
VOIP
Server
910
Internet
901
Cellular
Infrastructure
904
906
905
PC
9
0
7

Fig. 9

COOPERATIVE NETWORK FOR MOBILE INTERNET ACCESS

CROSS-REFERENCE TO RELATED APPLICATIONS

This patent does not claim the benefit of prior applications.

STATEMENT REGARDING FEDERALLY SPONSORED RESEARCH OR DEVELOPMENT

'Not Applicable'

REFERENCE TO A MICROFICHE APPENDIX

'Not Applicable'

TECHNICAL FIELD

The present invention relates to a computer method and system for connecting to a fixed network and, more particularly, to a method for connecting mobile users to the Internet.

BACKGROUND OF THE INVENTION

The Internet came into being in its current form in 1983 when all the computers connected to ARPANET—a military network—began using the same communication protocol. In 1986 ARPANET became NSFNET in a bid to link the USA's supercomputers. E-mail began to be commercially available in 1990 at much the same time that Mosaic—the first world-wide web browser—became a useful product. The Internet, comprising mainly the WWW (world wide web) and e-mail is now an extremely important business tool.

The world is now networked, much of this provided by TCP/IP (Transmission Control Protocol/Internet Protocol) but ATM (Asynchronous Transfer Mode) is still dominant in telecommunications. Most major companies have access to the Internet and the Internet backbone runs thru much of our environment. This infrastructure is made up from largely fixed, rigid connections; wires, routers, switches and similar.

Human beings tend to move around when engaged in work and leisure. An easy way to connect to the fixed network is needed for these mobile users. Mobile users are commonly envisaged as people using a cell phone or other radio device, but for our purposes a 'mobile user' is anyone away from their fixed home base.

It remains difficult for a mobile user to obtain a high bandwidth connection to the Internet for two reasons:

1. Wireless connections are inherently expensive as they use a rare resource—radio frequency spectrum—and require large infrastructure investments.
2. There are practical difficulties when away from home base, such as determining IP addresses, proxy server settings, negotiating billing and payment and security and privacy concerns.

Even the simple task of moving from one office to another within the same company can be fraught with complications and more often than not people fail to make connection when traveling away from home base. With so many companies relying on e-mail and the World Wide Web as a critical business tool, an inability to connect can prove costly to companies. In the last three years the ability to access the Internet has begun to migrate to mobile devices. Small mobile devices have tended to use a variety of specialist Internet access

methodologies with varying degrees of commercial success. They all suffer from the limitation of low bandwidth and high cost.

There are a number of methods by which users can get access to data from mobile wireless devices: SMS, HDML, WAP and I-Mode being the main standards. All of these standards suffer to some extent from problems such as limited bandwidth and complex authoring environments.

In SMS (short message service) users can send and receive simple, short text messages from their phone. A server at the mobile phone service either routes them to other mobile phone users or provides a gateway that translates the messages and sends them as e-mail to Internet e-mail services. The gateway will also translate incoming e-mail and send it to the user in SMS format. Although rudimentary, large successful businesses have been founded from sending these short messages demonstrating that people need data on the move and are prepared to pay for it.

In HDML (Handheld Device Markup Language) a web site is composed using specially modified tags. A wireless gateway translates these tags so that the mobile device may view them. A mobile device equipped with a micro browser capable of interpreting HDML can display this information on a small LCD.

In WAP (wireless application protocol) a web site is composed using WML (wireless markup language) and this information sent to a WAP gateway. The user gains access to this gateway either by dialing a RAS (remote access server) or by using a packet based system, commonly referred to as 'always on'.

In I-Mode specialist tags are again used to create pages formatted in compact-HTML. These are served over a gateway to users on a packet-based system.

In the above systems a specialist gateway is used to provide output formatted for mobile devices. An alternative method of accessing the Internet from a mobile device is provided by fixed wireless systems such as Bluetooth and the IEEE 802.11 wireless LAN standard.

In the 802.11 and Bluetooth standard two wireless devices establish a point to point or point to multi-point link using spread spectrum radio technology. The two wireless devices can be any type of electronic appliance—PC, PDA, Cell phone, microwave oven, home security system. This link replaces wires and does nothing to reformat the content of data.

Another wireless standard that has reached ubiquity in availability but has very little usage is IrDA (Infrared Data Association). In the IrDA standard two devices with IrDA capability positioned near one other can communicate using infra-red beams. The standard suffers from two problems. Firstly that the link is optical and therefore requires a clear line of sight. The devices must be positioned so that their ports are visible to each other or at least so that sufficient reflected light reaches the ports. Secondly the two computers need to have their IrDA ports configured and switched on. This is a non-trivial task: The hardware must be enabled—commonly power-saving and compatibility issues mean that computers are shipped with the port disabled. A device driver must be installed. Once the physical link is available a logical link is needed to allow file transfer or access to the hard disk of the companion computer. Security and privacy must be ensured.

Looking at problems of getting Internet access when mobile a number of steps are required.

1. Some form of driver software is needed to configure the user's hardware to allow a link to be made.
2. The physical link needs to be made. This may be by pointing the two computers at each other or connecting

a back-to-back USB cable, which has been provided by, for example, the hotel at which the user is staying.

3. The host user needs to enable and share certain services, such as printers and hard disks and network connections.
4. The host user needs, itself, to log onto the Internet.
5. The host user needs to act as a gateway for the connecting user, translating and forwarding packets onto the Internet.
6. The IP address of the host's interface needs to be configured to a non conflicting address with that of its Internet link.
7. The IP address of the connecting user needs to point to the host user.
8. The ports and proxy servers need to be set up. Even if no special set up is required a user who normally works in a corporate network with special settings will need to write these down and then delete them from the system, remembering to type them back in when returning to the corporate environment.
9. If the link has a cost to it a financial transaction needs to be entered into with metering and billing.
10. No simple solution is available to ensure a secure link.
11. E-mail may or may not require yet more steps to enable.

This multi-step process is fraught with difficulties and there are numerous potential points of failure. Most connections fail because either one of the steps cannot be completed, or each step is so time consuming that the user gives up before completing the task. Since mobile users are usually short of time, have limited resources (such as driver disks, spare cables) and have to configure multiple times as they travel thru different environments, the effort is very frustrating. Typically the above exercise is completed around 30% of the time with a mean time of effort of two and a half hours. Although many operating systems (such as Windows 2000™) allow connection sharing the implementation of these makes the task very complex.

An additional further complexity has been generated by the lack of interoperability between IEEE 802.11 and Bluetooth. Since both standards operate in the same frequency spectrum the two systems will tend to interference with each other.

Additional difficulties occur when a firewall is present. If a shared connection is obtained in the home or office, access to the Internet may be blocked or restricted by the activities of a firewall. The same technology that provides the firewall capability may also track usage and web sites visited providing a risk to privacy. Additionally information that is sent or received may be logged causing considerable security risk.

Although the cellular system provides some degree of roaming it provides no solution to inter-system roaming and uses expensive infrastructure, which results in high call cost. In addition the process of connecting a PC or PDA to a cell phone is complex and requires considerable reconfiguration of the device.

The current mode of access to the Internet is predominantly through a one-to-one commercial negotiation. Each person has a connection to the Internet via their own ISP. When visiting someone else's home obtaining access to the Internet is difficult. Calling the ISP requires another phone line. In the case of ADSL or cable systems the MAC address of the network adaptor is registered to the ISP so that a second user can't use the connection. You can install a network add enable connection sharing in your OS—for example Windows 2000™—but this involves a number of complex steps. In addition there is a significant security implications to this. Generally home and commercial networks are not set up with the anticipation that roaming users will be physically connected to the network inside the Firewall. Thus many network services are not secured against internal access. (It is not uncommon for a corporate network running Windows™ 2000 to have numerous hard disks shared without passwords.) Although the firewall blocks this from external access, a local connection would be inside the firewall.

Although these problems have existed for some time, the emergence of Bluetooth and IEEE 802.11 are encouraging people to connect more often.

While the above discussion is centered on the difficulty of human beings obtaining connection to the Internet it should be bourn in mind that autonomous machines have similar difficulties. Much talk is made of microwave ovens, video recorders and refrigerators talking together using home networks. It is also envisaged that automobiles will be similarly equipped. These systems are likely to become ubiquitous over the next five years as networking capabilities are built into appliances. These devices need to obtain services from somewhere and need to obtain a connection to the Internet. It will be prohibitively expensive to give each device it's own dedicated connection. The devices will also need services tailored to them. One can consider that an appliance once manufactured an shipped becomes a roaming device in need of connection to the Internet.

SUMMARY OF THE INVENTION

An embodiment of the present invention provides a method and system for connecting to a network—typically the Internet via a cooperative membership arrangement. Members of the cooperative networking service have an identifier that uniquely identifies them as a member of the service. A fixed member (host) enables its link capabilities for mobile users (clients) but directs those links to communicate with a secure piece of software—the Cooperative Tunneling Agent (CTA). Upon access by a mobile member, the CTA sends both identifiers and status information to a server. The server contains preference information on all members that can be referenced by their identifiers. The status information is compared to the preferences of the two members and, if the comparison is successful, a link to the Internet is provided to the mobile member. If the comparison fails the members may be notified and allowed to modify the preferences, make a one-off exception or reject the link. The CTA can work with or without reference to a server. When working independently of the server the comparison is made using data from the host and the client and the server is only accessed if additional information or service is required.

The CTA ensures that the mobile user has no access to the fixed user's computer other than as a pipe thru to the Internet. To ensure that the mobile user's security and privacy are protected the data packets may be encrypted and sent via the fixed host to the server. The server may decrypts the data and sends it to the Internet. Return data is similarly dealt with. By this method the fixed member cannot eavesdrop on the mobile member's data or voice traffic. Such encryption can be set at a variety of strengths according to user preference and applied selectively depending on the data type. Routing of the data may also be performed selectively according to data type.

BENEFITS OF THE INVENTION (O&A)

An advantage of the system is that two members can quickly and simply set up secure link so that one member can access the Internet thru the other members network connection without causing security risks to either member.

A further advantage of the system is that one member can use the low cost access method available to another user.

A further advantage is that voice calls can be sent to the host and routed over the IP network using VOIP technology thus dramatically reducing costs.

A further advantage is that voice calls can be routed thru the cellular system to a local VOIP server thus saving long distance cellular charges.

A further advantage is that by simplifying the operation of logging onto the network thru fixed nodes commercial node providers will find it easier to acquire customers and the service to those customers will be improved and simplified.

A further advantage is that by providing a universal logical standard for connection incompatible transport protocols can be made compatible. In the example of Bluetooth and IEEE 802.11 the CTA can ensure that the protocols synchronize to divide their spectrum usage in either time domain or frequency domain for all members present.

A further advantage of the system is that since access is automated according to pre-set preferences the system can be used for the connection of autonomous appliances and machines who are not able to make complex decisions regarding optimal network access.

The link can transport either data or voice. In the case of voice the CTA handles the VOIP (voice over IP) protocol. VOIP and Multimedia can be either sent via the complete path thru our secure server or split apart to take a more optimum routing.

The software allows the user to access the Internet, send and receive e-mail and obtain high bandwidth services such as MP3 files and movies. The network is cooperative as each user agrees to both give and receive access depending on their status. Each user is a node. A cell phone located within 100 feet of a fixed host device can connect to the Internet through that device, obtaining phone calls at a fraction of the cost of a regular cellular connection. Similarly a portable computer or PDA would be able to connect through a host node and obtain a high bandwidth link to the Internet for a modest sum of money. It is anticipated that there will be two main types of connection; cooperative connection and commercial connection. In a cooperative connection, a fixed user connected to the Internet provides a bridge so that other mobile users may access the Internet through their link. This link is provided free of charge in return for reciprocal roaming rights. In a commercial connection the user locates a commercial node and the CTA negotiates connection and payment terms according to preset user preferences.

While this invention is primarily targeted at simplifying the task of connecting using fixed wireless devices such as Bluetooth or 802.11 it will apply to any connection methodology—IrDA port, network cable, serial cable, USB crossover connector et. al. Our invention is intended to facilitate connection to any rigid network infrastructure but the Internet is the most important of these.

Means for cleanly handing off connection from one system to another are provided by the CTA. By storing both the customer's Cellular provider information and the Cooperative network ID the call can be routed through either system. In addition if a voice call is in progress and a Bluetooth cooperative connection becomes available the call can be set up in the VOIP system and handed over and visa versa.

The technology provides the function of a bandwidth moderator. As a user moves about their daily life from home to office via different modes of transport, their requirements for Internet connection change in characteristic, and the mode of connection changes. Our technology sits between the user and the Internet constantly negotiating the most cost effective means by which they can gain access. While today the modes seem simple—it's a wired or a wireless link—this will not remain so in the future and all software systems will require an agent to perform this task. It is an objective of the present invention to optimize the 'last mile'.

This system differs substantially from the cellular system in that a substantial amount of network hardware is provided by the members of the network and not by high capital value dedicated infrastructure. It is generally a surprise to people when they realize how often they are within 100 feet of an Internet connection. Additionally the cellular system provides no handoff methodology between differing classes of carrier.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. **1** illustrates tunneling Internet access in one embodiment of the present invention.

FIG. **2** is a flowchart of the process by which a link is established.

FIG. **3** is a table showing the data record structure for a customer.

FIG. **4** is a flowchart of the process by which cooperative networking is enabled.

FIG. **5** is a diagram illustrating recursive service discovery.

FIG. **6** is a block diagram of the elements of the system.

FIG. **7** is a table describing a day in the life of a user.

FIG. **8** is a flowchart of the matching algorithm.

FIG. **9** is a diagram illustrating voice over IP and service handoff.

DETAILED DESCRIPTION OF THE INVENTION

The present invention provides a method and system for easy, secure, mobile Internet access thru fixed connections points These fixed connection points being computers that are connected to the Internet and have another port available for connection by one or more mobile users. A fixed connection point can itself be a portable device, such as a portable PC, and may be moved from time to time. The term is a relative one. To a mobile user a PC with an Internet connection is a fixed device. For the purpose of discussions regarding the system operation these fixed points will be described, as hosts and a mobile user will be called a client. The host can be any form of general purpose PC or a custom built device, which as two connection methodologies and an ability to selectively connect those methodologies under program control.

The simplest system has a host and a client. The host will generally communicate with a server to transfer data and receive services but a host can run in stand-alone mode if connection to a server is unavailable or if client security and privacy are not high concerns such as when the host and client are used by the same person.

The system is composed of three main parts: The software application, the web server and associated web site and specialist network services such as streaming and VOIP which are provided by other parties.

The Link Functions Include:

1. Download of software.
2. Installation of software.
3. Updating of software.
4. Registration of the user to the service.
5. Testing the system.
6. Location of another wireless node or enabled device.
7. Backup location of any Node. IrDA, wired, dialer . . . .
8. Determination of payment conditions: cooperative free, cooperative with cost, commercial.
9. Determining whether the user permission matches the link available.

10. If it does, establishing the link.
11. If it does not, notifying the user of the availability and giving them the option to accept or reject plus update their user profile.
12. Provide voice over IP.
13. Call handover to cellular network
14. Data handover to cellular network.
15. Encrypt/Encode data for safe sending to the remote web server.
16. Location of VOIP network nodes (Akamai™, Lipstream™ et. al.) and direct communication with streaming servers.
17. Billing, most recent call information.
18. Fault logging and diagnostics.

The Tunneling Functions Include:

1. Identify fixed line connection or another wireless node.
2. Connect to cooperative server and register.
3. Activate and monitor cooperative connection and enumerate service.
4. Negotiate link with request from cooperative member.
5. Provide secure link (tunnel) to Internet.
6. Handover to alternate link upon request or change in network status.

The Server Functions Include:

1. Registration of users.
2. User preference setup.
3. Service list/Portal to customer services.
4. Decoding/De-encrypting transmission and forwarding to web.
5. Provide an Internet store and forward
6. Negotiation of streaming links
7. Negotiation of VOIP link
8. Allow registration to partner services.
9. Negotiate handover upon network status change.

Third Party Functions that May be Linked to Include:

1. Provide VOIP service (Lipstream™, Net2phone™, Akamai™)
2. Provide Streaming service (RealNetworks™, Akamai™, Microsoft™, QuickTime™)

The software application runs on both the client and the host, varying its mode of operation depending on its function in the network. (It is possible for the software to run within a browser thus giving a 'software-less' mode of operation. Is will be obvious however that software is indeed running in this instance.)

The web server provides the user interface to the service, allowing people to register and also the configuration software to correctly set up each machine and ensure that the software application which runs on the client's and host's is up-to-date and correctly configured.

In the case of multi-media and voice these IP packets may not be routed to the server but may travel via a third party routing service for optimum performance.

The preferred embodiment of the Invention will now be described with the aid of figures. The embodiment assumes that a short-range wireless link such as Bluetooth™ will be available, and the reader is directed to the Bluetooth specification for further implementation detail. In addition a user skilled in the art should be able to translate the flowcharts into useable code by reference to examples provided in the literature, for example the Microsoft Developer Network™ for the Windows™ operating systems.

FIG. **1** illustrates the general layout of the network. A roaming device, **106**, which might be a mobile computer, PDA, cellular telephone or home appliance, comes within range of the fixed wireless link of a member, circles **107-110**. The mobile device determines the connection methodologies available to it and their relative merits and then connects to the host using the best available standards.

Once the link has been established the cooperative tunneling agent examines the preferences of the roaming member and the fixed member and compares these with a database stored on the server. If the comparison generates a match then standard IP packets are encrypted and sent from the roaming device **106**, thru the fixed device **104** or **105** thru any present firewall **103** thru the Internet **102** to our server via a link to the Internet **111**. Our server **101** decrypts the packets and sends them onto the Internet via the link. Return packets from the Internet are passed back to the server and are similarly routed back to the member via the encrypted links.

There are two scenarios; corporate and home users. A home user depicted as PC labeled **104** typically makes a direct connection to the Internet **102** without the benefit of a firewall **103**. A corporate user connects to the Internet thru a proxy server and firewall **103**. These two functions are shown as a single block but may be comprised of several physical units; a router, a server, a proxy server, a firewall and a security monitoring computer. For our purposes it is not important how many physical or logical blocks are involved merely that there is a Firewall function that we need to negotiate. Firewalls provide a range of services. Primarily they block all ports except the commonly used business ports such as http port **80**. This stops hackers accessing systems using older methodologies that often grant considerable powers to a user. In addition the firewall may block specific web sites thru the exclusion of certain IP address lists—'black list'.

These preferences may be appropriate for the corporation but wholly inappropriate for the member requesting the tunnel. Therefore the tunneling agent can encode all packets and sends them thru port **80** bypassing any corporate specific snooping of the data. If the UDP ("user datagram protocol") port is available this may be preferentially used for sending VOIP and other real time packets.

References to the server in this description may apply to any server providing our service. There may be more than one server on the network. Accesses may be cached at any point in the network. There may be several synchronized servers located geographically to improve throughput and efficiency of the system. The actions of the server may be split across several logical or physical devices so as to optimize performance. In addition some specialized tasks may be offloaded to third party servers. The server may be the computer of another member such that there are no specialist servers in the system and all tasks are distributed. In such a case the members database might be duplicated and distributed thru the network to provide security and redundancy and avoid a single point of failure.

FIG. **2** is a flowchart giving the steps required to establish a link and obtain access to the Internet. The tunneling agent running on the client computer sits in a loop attempting to find a network node **201**. The polling rate can be set by user preference or turned off and re started by user command at a later stage. In the case of a Bluetooth™ radio link the link protocol will find a network node and establish the physical link in step **202**. At this point a physical link exists between the two devices but no services are yet available The host computer has been pre-configured to direct all physical links to the CTA (cooperative tunneling agent) which acts as a gate keeper for all network access thus the logical link **203** is set up to point only to the CTA. The client then sends the client ID and associated information to the CTA. If this is the first access by a user to the host system the local match step **205** will always fail. However the match can fail for two reasons. Firstly that the preferences don't match immediate failure can

be notified to the client without reference to the server. If the preferences match but insufficient information is held to determine whether a complete link can be made the host will contact the server. (In certain circumstances such as the request for a non-client-secure link the client and host can operated independently of a server.) The client and host information is sent to the server in step **206** and the server attempts to make a match in **207**. If the match fails a link is not made and the user is notified and given options to modify their preferences and start again. If the match is successful the link is made and link information is cached on the host machine. This information can be used to reestablish the link without reference to the server if the link is broken. A mobile user who is on the edge of range may be constantly moving in an out of range so caching the link information allows the link to be made and broken quickly without a server request being required every time.

FIG. **3** is an excerpt of a member record showing the important elements in creating a match between cooperative members. A person skilled in the art will appreciate that there will be other fields required to fully describe the customer and log historical information but these are generic to all ISP (Internet Service Provider) customer records. Remembering that every user can be a host or client, the record is appropriate for all members. The record is divided into three main sections: general data **301**, tunnel provision **302** and tunnel request **303**. In simple terms the member agrees to grant access to the Internet on certain terms to other members. Members define the parameters on which they will accept access. If two parties come into range and the granting and accepting criterion match, a link is enabled. The most common link is a fixed member (host) connected to the Internet by an always-on connection, such as an ADSL modem, granting free access to a mobile member. Another common connection occurs where the host is connected by metered access. The host can inform the client of the cost prior to making a connection to the network. The process by which this information is used to establish a link is detailed in FIG. **8**.

FIG. **4** is the flow diagram by which cooperative networking capability is enabled. In step **401** the user logs onto the website to requests the service. In step **402** software is downloaded to the member device. In step **403** the software scans the members system and determines the connection methodologies available, sets parameters and directs the flow of data from those methods to a secure piece of software. In step **404** the user sets the tunneling preferences that they wish to grant to other members along with their personal details. In step **405** the user sets the access provision preferences they would like when they use the network. In step **406** the system summarizes the preferences and confirms they meet the customer policies pertaining at the time. These preferences are stored on the host/client with the associated user ID. Confirmation of credit approval can be given at this point.

One skilled in the art would appreciate that the software could be provided by other means such as a floppy disk or CD-ROM.

Rather than downloading and running the software on the client system, the software could run on the server and client system using a method such as Active Java™ Pages. However in order to configure the client system the software will need to undertake a number of low level tasks which might be protected from modification by a Java program and downloadable program elements are generally required.

Some operating systems, when connected to a new network, automatically log onto a default network specific web page upon boot-up. This feature can be readily provided so that a client system has only to connect their computer and turn it on and the host node will automate steps **401** and **402**

In the absence of a network connection the host computer may store a copy of the installation and registration software so that the registration and setting of preferences can be performed off-line and then uploaded to the server upon next connection.

The configuration software may be 'beamed' from one user to another or transferred over the link to user who does not yet have the service by a push or pull methodology.

FIG. **5** illustrates recursive service discovery. In this specification we have generally discussed systems where there is one fixed node connected to the Internet providing one or more Internet connections to mobile nodes. However it may often happen that a mobile node is not within range of a fixed node but is within range of another mobile node that is itself within range of a fixed node: a two hop connection. In this figure we show a three-hop connection. A mobile user **505** wants to gain access to the Internet. Thru the wireless network it connects to another mobile user **504** and requests a tunnel to the Internet. This unit is not connected to the Internet thru a wired connection and so itself makes a request to another unit **503**. This unit, although fixed, has no hardwired Internet connection and so connects another Computer **502**, which does have a connection to the Internet **501**, and a link is made. In this example the link is established on a request basis as it is undesirable to establish a permanent link for power conservation: a pull system. However the link between **502** and **503** can be established once and be substantially permanent as both units have mains power supplies: a push system Thus the network can spread out to all permanently powered computers that are within range of each other and can be further increased in size on an ad hoc basis by requests from mobile units. The scope of the network may be pushed outwards to encompass all available nodes or pulled outward by devices wanting connection to the fixed network.

In order to save power it may be necessary for systems to synchronize their waking by registering a time pulse from the server and waking at predetermined times. (Packet delay time must be measured and added to the time synchronization pulse to ensure accurate synchronization of the servers.) Thus all the components in a system may wake at a predetermined time to attempt to push the network outwards.

FIG. **6** illustrates in block diagram form the elements of a system. The client Browser **602** is pointed at the cooperative network website. It downloads and runs a program that sets up the computer. (This set-up was described in detail in association with FIG. **4**.) The set-up means ensures that the correct drivers are installed on the computer to physically utilize the ports of the computer and that the IP stack and network resources are available and can be intercepted. Depending on the exact O/S, the CTA cooperative tunneling agent will either redirect all network calls thru its routines or provide a simple means for switching itself in when needed. The best method is usually described with examples in the documentation of the operating system. The software for the client is the same as that of a host system but the CTA on a host system only needs to talk to the uplink as there is no further downlink to manage. That said the downlink must be left in a state awaiting connection so that a multi-hop request can be made.

The host computer runs two IP stacks and two sets of network drivers. The first set **606** and **609** are used to communicate with the client. The second set **608** and **610** are used to talk to the Internet. The CTA **607** provides a low level link between the two. The host browser and applications are generally unaware of the presence of the link unless the host desires to know, in which case it can either run a utility, point

to a pre-defined web page or enable a status notification. The host applications utilize the IP stack and drivers **608** and **610** in the normal way. The server comprises a server engine **617** that performs all the tasks of linking with the Internet and decrypting and re-routing packets. A store containing the latest configuration software **615**. The matching rule file **616**. The member database **614** and the web pages which together form the web site **613**. A packet front the client takes the following path. In the case where the client makes a web page request the browser routes the call to the IP stack in the client computer. The CTA intercepts the request, encrypts it and encapsulates the packet in a new packet, which is routed to the host. The host picks up the packet and routes it thru the CTA to the outgoing stack, adjusting the IP address appropriately. The server receives the packet, de-encapsulates it, decrypts it and sends it onto the web. When a reply packet arrives, it is encrypted, encapsulated and sent to the client. Further details on implementation of tunneling protocols can be found in the literature.

FIG. **7** is a table describing one potential typical day for a mobile Internet user and illustrates the complex options available for connection and the many hand offs between systems. Starting in the home **701** the user has a cable modem giving access to the Internet at 500K bytes per second. Walking to the train **702** potential access speed drops to 9.6 Kbps using their mobile phone. On the fast moving transport mode of a train **703** the potential connection speed is limited, even in the new 3G networks that are being proposed. Once at work a free connection is available to the Internet using the office LAN **704**. When the user goes to a meeting they disconnect from their LAN and move either to a different LAN node or a mobile network provided by the IEEE 802.11 standard **705**. As they go out to lunch in a local coffee bar Starbucks™ they have access to a public Bluetooth node **706**. Back in the office **707** LAN access is again available. Cell phone is available as they walk to their car **708** and drive in their car **709**. At a gas station where they stop, a Bluetooth link is again available for high speed access **710**. At a friend's house an ADSL link is available **711**, but connecting to that LAN could be by a variety of means, including Bluetooth link or network cable. After leaving their friends house our user goes via Taxi **712** to a Hotel where a high speed Internet link kit is available in the closet of their room **713**. It is an object of our Invention to make such mobile use simpler by provision of means to mediate the handoffs between these systems, store user and network data centrally and to provide common access links to all these different networks.

FIG. **8** is a flowchart illustrating the matching process. The matching process obtains the host **801** and client ID **802** and looks up the record on the database **803**. It then attempts to match the access permission with the desired access **804**/**805**. If this fails the client is asked whether it wants to change preferences **806**. (Typically the failure will occur because the client has no settings for obtaining a paid link to the Internet if the host is a commercial link or a cooperative link with non-zero incremental cost for access. The host will want to pass this cost onto the client.) This will require the client to amend their record **807** to provide billing info and to set the maximum connection charge and per minute charge that they are prepared to pay for an Internet connection. In the case of a paid link they can further set a budget for link costs.

FIG. **9** illustrates handoff between a fixed wireless, Internet based, VOW system and the cellular system. A mobile user **907** is within range of two methods for placing a call; a PC running our cooperative networking service and a cellular tower. The call might be preferentially placed to either unit based on the users pre-set preferences or based on the current situation. Both links could be made or broken independently and exist simultaneously. In the case of connection made over the Internet voice packets are sent over the air using a wireless link such as Bluetooth or IEEE802.11 to the host **905** These packets are routed thru the Internet **901** to a VOW server **903**. The VOW server converts IP packets to a form suitable for use over the PSTN and handles making and breaking the connection to users. There are many VOW servers, two are shown here, located in each region such that the call is placed at a local rate. If the user moves away from the fixed wireless link **905** or the Internet becomes congested such that call is no longer of sufficient quality, a factor that can be determined either automatically or by the user the call can be dialed on the cellular link. There are a number of methods that handoff could take. Firstly the call could be dropped by the VOW server and a new call initiated end-to-end thru the cellular network to the destination **908**. This would result in a break in the call. More suitably the mobile device would call a VOIP, either the existing server **903** or a server located near to the caller **910**. The VOW server would take information from the cooperative network server and route the call to the already established PSTN link and then to the destination **908** thus maintaining the call. A benefit of this is that the call is not dropped, interrupted or reinitialized. A further benefit is that mobile device can place a call to a local VOIP server **910** which forwards the information to a VOW server local to the receiving party thus saving cost. The call routing between the two VOW servers can be via the public Internet (dashed line) or thru a proprietary network.

The host node may send data using IP over the air or it can emulated the Bluetooth™ air interface and appear as a cordless telephone base station for the client device. By this method no specialist software needs to run on the client device if that device is a handheld phone or similar. Security can be preserves by not decrypting the air interface on the host system and passing the un-decrypted information onto the server.

RAMIFICATIONS

The use of this technology will enable a number of commercial services that have hitherto been prohibitively expensive or complex.

Roaming users will be able to get automatic high bandwidth connections to the Internet wherever there is another member present. Since members can be commercial or cooperative in their action this will allow ubiquity of service.

One can envisage a member parking their car outside their home. The CTA connecting automatically to their home computer and synchronizing their MP3 music collection with their on board hard drive.

One can envisage a member driving into a gas station, purchasing a movie for their children using the onboard computer and while gas is being pumped and paid for the movie is downloaded to the on-board entertainment system.

The invention claimed is:

1. A method of connecting a client system to a target network via a host system that has link capabilities for connecting to the target network comprising:

providing application software to a client system so the client system may act as a host or a client; said application software being configured on said system that identifies said system as a member of a cooperative networking group;

connecting the client system to the host system via application software on a host system; said host application software being configured to communicate with the application software on the client system that has the configured client application software that identifies the

client system as a member of the cooperative networking group; said host system being capable of accessing the target network;

linking the client system to the target network via a tunnel through the host system; and

preventing the client system from accessing resources of the host system by the steps of:

enabling the host system's link capabilities for clients but directing those links to communicate with the host application software;

causing the client application software to intercept the communications of the client system directed to the target network, encapsulating them in a new packet and sending them to the host system; wherein all packets sent by the client application software is directed to the host application software.

2. The method of claim **1** further comprising the steps of

determining requirements for connecting the client system to the target network; and

linking the client system to the target network based on the requirements; wherein the determining step is made in a server.

3. The method of claim **2** where the determining step further comprises:

determining a cost of obtaining a link; and

comparing the cost with a predetermined limit stored by said client system where a match will be found if the cost is less than or equal to the predetermined limit.

4. The method of claim **1** wherein the target network is the Internet.

5. The method of claim **1** wherein the target network is a public network.

6. The method of claim **1** wherein the target network is a PSTN.

7. The method of claim **1** wherein the client system is a mobile phone.

8. The method of claim **1** wherein the host system and the client system are members of the cooperative networking group that gives reciprocal roaming rights.

9. The method of claim **1** wherein the host system is not connected to the target network and will make the connection to the target network upon request by the client system.

10. The method of claim **1** wherein any data sent by the client system is encrypted prior to being sent to the target network such that the privacy and security of the client are maintained.

11. The method of claim **10** wherein the encryption can be selectively applied to the data depending on preferences.

12. The method of claim **1** wherein the connecting step comprises using a wireless connection technology.

13. The method of claim **1** wherein the linking step occurs automatically once the requirements for connecting the client system to the target network have been determined.

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